Customer Behaviour Analysis

About the Dataset

- **Type:** Retail / E-commerce Dataset
- **Nature:** Structured (Tabular)
- **Level:** Customer Transaction Level
- **Data Mix:**
  - Numerical → Age, Purchase Amount, Previous Purchases
  - Categorical → Category, Payment Method, Season

This dataset is used for **customer behavior analysis + business decision-making**

Domain Background

In the **e-commerce industry**, companies track every customer interaction:

- What customers buy
- How often do they buy
- How much do they spend
- Whether they respond to discounts

Why This Data Is Important

This data helps businesses:

- Identify **high-value customers**
- Understand **purchase patterns**
- Optimize **discount strategies**
- Improve **marketing campaigns**
- Increase **customer lifetime value (CLV)**

Business Problem

How can a retail company understand customer purchasing behavior to increase revenue, improve customer retention, and optimize marketing strategies?

Breaking Down the Problem

Low Customer Retention

- Customers are not purchasing frequently

Need to understand:

- Who are frequent buyers?
- Who are one-time buyers?

Ineffective Marketing Campaigns

- Discounts and promo codes may not be effective

Questions:

- Do discounts increase purchases?
- Which customers respond to promo codes?

Product Demand Understanding

- Which category performs best?
- Which products generate the highest revenue?

Payment Behavior Insights

- Which payment methods are preferred?
- Are digital payments increasing?

Seasonal Demand Variation

- Does season impact sales?
- Which season drives maximum purchases?

Customer Segmentation Problem

Need to identify:

- High-value customers
- Frequent buyers
- Discount-sensitive customers

Data Dictionary

Column Name	Description
Customer ID	Unique identifier for each customer
Age	Age of the customer
Gender	Gender of the customer (Male/Female/Other)
Item Purchased	Specific product purchased
Category	Product category (Clothing, Electronics, etc.)
Purchase Amount	Amount spent by the customer
Location	Customer's geographic location
Size	Product size (S, M, L, etc.)
Color	Product color
Season	Season of purchase (Winter, Summer, etc.)
Review Rating	Customer rating for the product
Subscription Status	Whether customer is subscribed (Yes/No), Subscription Program
Shipping Type	Type of shipping selected
Discount Applied	Whether discount was applied
Previous Purchases	Number of past purchases
Payment Method	Mode of payment
Frequency of Purchases	Purchase frequency (Weekly, Monthly, etc.)

EDA- (Exploratory Data Analysis using Python)

Data Loading

What you did:

- Imported **pandas**
- Loaded dataset using:

```
#importing the dataframe/dataset
df=pd.read_csv('customer_shopping_behavior.csv')
```

Why did you do it:

- To bring raw data into a structured DataFrame
- Enables further analysis and transformations

```
df.head()
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method
0	2701	22	Female	T-shirt	Clothing	68.0	California	XL	Olive	Winter	3.2	No	Standard	No	No	36.0	Cash
1	521	51	Male	Sunglasses	Accessories	84.0	South Carolina	M	White	Spring	3.9	Yes	Free Shipping	Yes	Yes	20.0	Debit Card
2	3157	18	Female	Shirt	Clothing	50.0	Montana	M	Black	Winter	3.1	No	2-Day Shipping	No	No	18.0	Cash
3	1687	22	Male	Gloves	Accessories	75.0	Illinois	L	Red	Fall	4.2	No	Store Pickup	No	No	25.0	Cash
4	2929	40	Female	Jewelry	Accessories	80.0	Alabama	L	Yellow	Spring	3.6	No	Store Pickup	No	No	17.0	Credit Card

Initial Data Exploration

What you did:

- Viewed data using:
  - df.head()
  - df.tail()
- Checked dataset summary:
  - df.describe(include='all')
  - df.info()
- Checked missing values:
  - df.isnull().sum()

Why did you do it?

- To understand:
  - Data structure
  - Data types
  - Missing values
- This step helps identify **data quality issues early**

Data Consistency Check (Category Correction)

What you did:

- Checked unique values in:
  - Item Purchased
  - Category
- Found inconsistencies between **Item Purchased and Category**
- Created a **mapping dictionary**
- Updated category using:

Why did you do it?

- Categories were inconsistent or incorrect
- Ensures:
  - Accurate grouping
  - Correct business insights

This is a **critical real-world data cleaning step**

Handling Missing Values (Size Column)

What you did:

Handled missing values based on category:

- Electronics → "Not Applicable."
- Accessories → "Not Applicable."
- Footwear → "Free Size."
- Clothing → Filled with **mode**

Why did you do it?

- Different categories require **different business logic**
- Example:
  - Electronics don’t have size → Not Applicable
  - Clothing → size is important → use mode

This shows **domain understanding**

Handling Missing Values (Review Rating)

What you did:

- Checked:
  - Mean, median, mode
- Filled missing values using:
- Rounded values

Why did you do it:

- Instead of the global average, the **product-level mean is used**
- This keeps ratings **realistic and meaningful**

This is an **advanced imputation technique**

Handling Missing Values (Purchase Amount)

What you did: Replaced with the mean value.

Why did you do it?

- Maintains **product-level pricing consistency**
- Avoids distortion in revenue analysis

Handling Missing Values (Previous Purchases)

What you did: Replaced with the 0 (zero) value.

Why did you do it:

- Missing values imply **no previous purchase**
- Important for:
  - Customer segmentation
  - Loyalty analysis

Duplicate Handling

What you did:

- Checked duplicates:
- Identified duplicates using Customer ID
- Removed duplicates:

Why did you do it?

- Ensures:
  - Each customer appears only once
  - No double-counting in analysis

Critical for **accurate reporting**

Column Standardization

What you did:

- Converted column names:

```
df.columns = df.columns.str.replace(" ","_")
df.columns = df.columns.str.lower()
```

- Renamed column:

Renaming the Headers

```
df.columns=df.columns.str.replace(" ","_")
df.columns=df.columns.str.lower()
```

Why did you do it?

- Makes column names:
  - Clean
  - SQL-friendly
  - Easy to use in Power BI

Data Export to SQL Server

What you did:

- Installed required libraries:

```
pip install pyodbc sqlalchemy
```

- Connected to SQL Server
- Exported data:

Why did you do it?

- To move cleaned data into **SQL for further analysis**

- Enables:
  - Query-based insights
  - Integration with BI tools.

Data Analysis using SQL Server

1. Which category generates the highest revenue?

	category	total_revenue
1	Electronics	486366.96
2	Accessories	194866.87
3	Clothing	183188.39
4	Footwear	111236.2
5	Outerwear	18524

2. Are discounts actually increasing purchase value?

discount_applied	total_revenue	Avg_Purchase
No	502241.64	182.24
Yes	491940.78	219.22

3. What is the total revenue generated by male and female customers?

gender	total_revenue
Male	390964.526238359
Female	352060.29314048 5
Other	251157.60467728

4. **Which customer used a discount but still spent more than the average purchase amount?**

	customer_id	purchase_amount	discount_applied
1	4568	276.343789954338	Yes
2	4256	734.47734939759	Yes
3	4289	1282.44	Yes
4	4576	831.344565217391	Yes
5	4319	685.231470588235	Yes
6	4708	735.773625	Yes
7	4584	685.231470588235	Yes
8	4607	557.24	Yes
9	4851	685.231470588235	Yes
10	4845	692.11	Yes

5. **Which are the top/bottom 5 products with the highest average review rating?**

	item_purchased	avg_review_rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.8
5	Skirt	3.79

	item_purchased	avg_review_rating
1	Phone	2.94
2	Headphones	3.08
3	Watch	3.1
4	Bag	3.17
5	Laptop	3.19

6. **Average purchase: Standard vs Express shipping**

shipping_type	total_revenue	total_orders
Express	358.12	1199
Standard	340.57	1201

7. **Do subscribed customers spend more? Compare average speed and total revenue between subscribers and non-subscribers.**

subscription_status	total_customers	total_revenue	avg_spend
Yes	1579	427245.4	270.58
No	3421	566937.02	165.72

8. **Top 5 products with highest discount usage %**

item_purchased	total_number_of_times_sold	Number_of_times_sold_when_disocunt_applied	discount_percent
Laptop	143	74	51.748251748251
Phone	171	88	51.461988304093
Headphones	166	85	51.204819277108
Hat	154	77	50.000000000000
Watch	157	78	49.681528662420

9. **Segment customers into new, returning, and loyal based on their total number of previous purchases, and show the count of each segment.**

customer_segment	Customer_Count
Loyal Customers	3085
New Customer	561
Returning Customer	1354

10. **What are the top 3 most purchased products within each category ?**

category	item_purchased	most_purchased	RNK
Accessories	Jewelry	171	1
Accessories	Bag	164	2
Accessories	Belt	161	3
Accessories	Sunglasses	161	3
Clothing	Shirt	315	1
Clothing	Blouse	171	2
Clothing	Pants	171	2
Electronics	Phone	171	1
Electronics	Headphones	166	2
Electronics	Watch	157	3
Footwear	Shoes	303	1
Footwear	Sandals	160	2
Footwear	Sneakers	145	3
Outerwear	Jacket	163	1
Outerwear	Coat	161	2

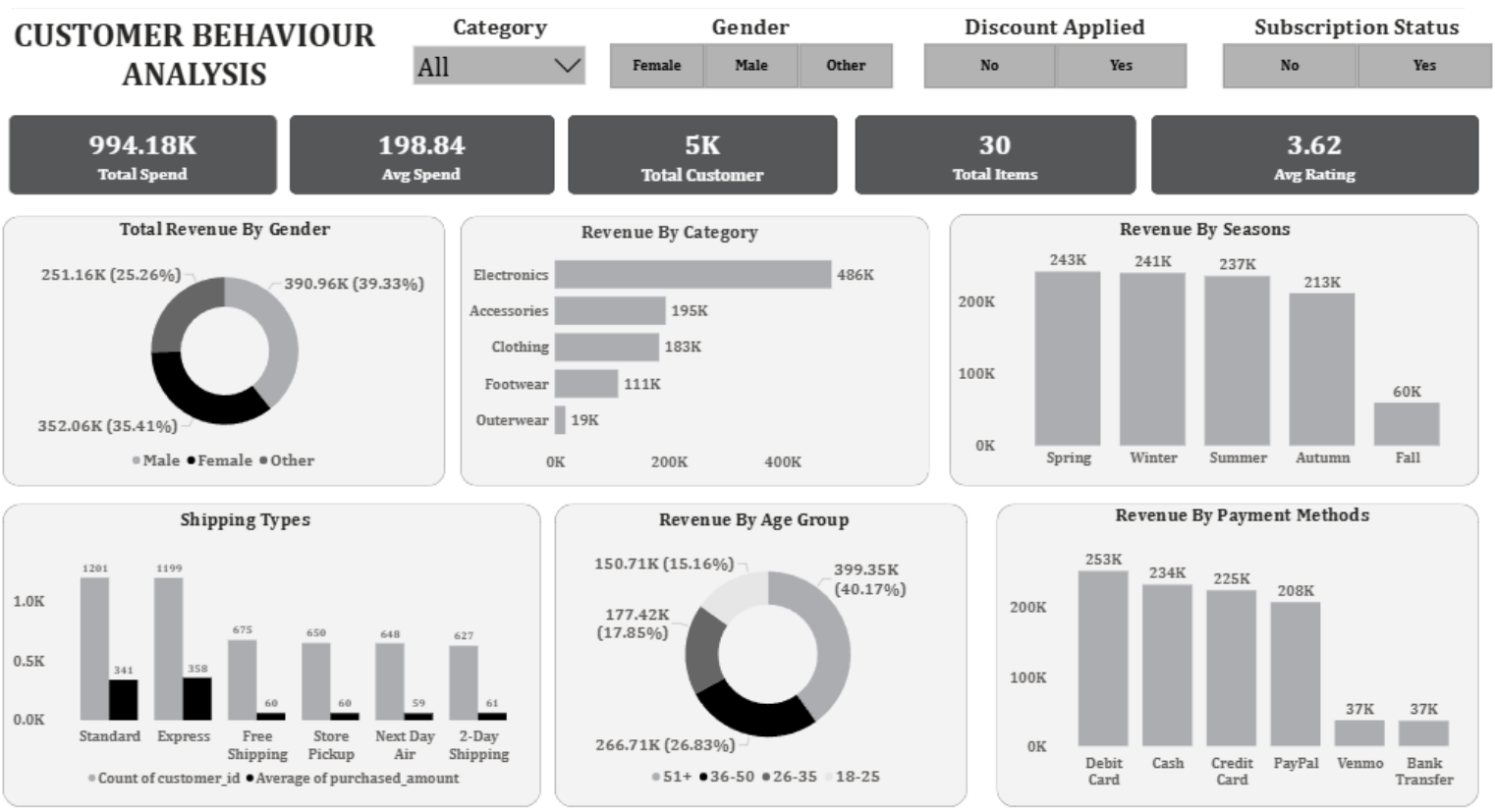
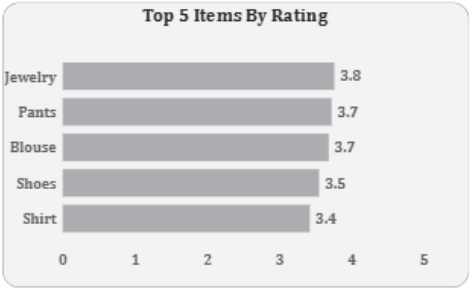
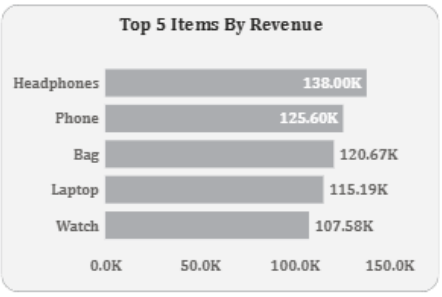
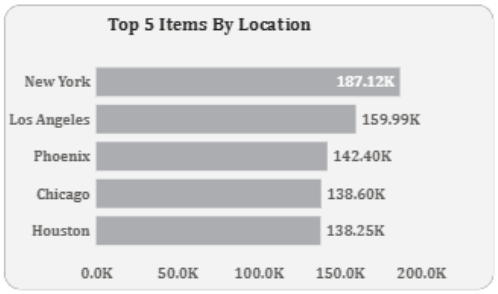
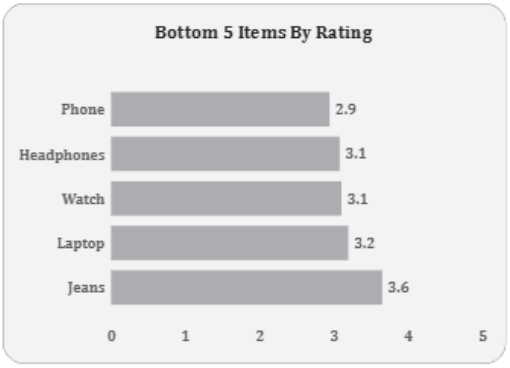
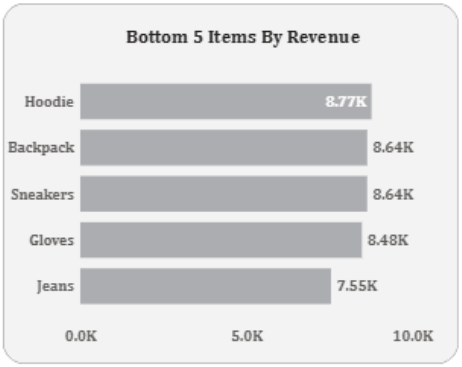
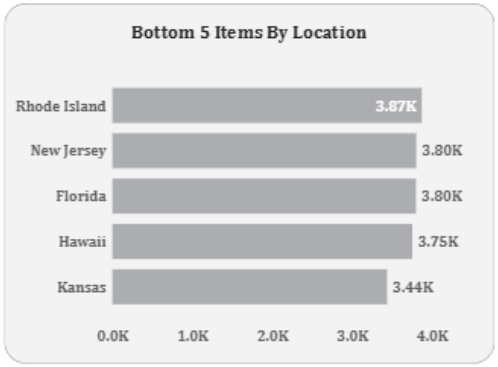
11. Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

Customer_type	subscription_status	Customer_count
Repeat Buyers	No	2763
Normal Buyers	Yes	389
Normal Buyers	No	658
Repeat Buyers	Yes	1190

12. **What is the revenue contribution of each age group?**

age_group	total_revenue
51+	399345.32
35-50	266706.7
26-35	177421.93
18-25	150708.48

Power BI Dashboard



994.18K	198.84	5K	30	3.62
Total Spend	Avg Spend	Total Customer	Total Items	Avg Rating